East West University

Task1 Report

Course Code:	CSE 475
Course Name:	Machine Learning
Section:	01
Dataset:	https://ninapro.hevs.ch/instructions/DB2.html
Title:	A Comparative Study on Graph Construction Strategies for Hand Gesture Recognition based on sEMG Data with Graph Neural Networks.
Semester:	Fall 2026

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Submission date

16th October, 2026

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Kaggle Link:
https://www.kaggle.com/code/marjan15/group05-ninaprodb2-task1-eda
GitHub Link:
https://github.com/Marjan15H/Group05_NINAPRODB2_CSE475

1. Introduction

Surface electromyography (sEMG) is the process of measuring the electrical activity of muscles with electrodes placed on the skin. Since each hand and wrist motion engages different muscles, Recognise gestures, such as in prosthetic hands and human-computer interaction. The problem is that the signals are noisy and the variations are very large, between The signal is a mixture of people and gestures.

NinaPro DB2 is a public sEMG benchmark including 40 subjects, 12 electrodes and 49 hand and wrist data sets. gestures. The 12 electrodes are located at various points on the forearm and upper arm, so that Relationships can be represented as a graph (nodes are electrodes, links between nodes are the edges). This is why the Graph Neural Network (GNN) track suits this data set: A GNN can leverage the relation to learn Between channels and not one channel alone.

2. Objectives

The aim of Task 1 is to understand the data and the existing research well enough to choose a sound approach for the later tasks. Specifically, we want to:

• Describe NinaPro DB2 (subjects, channels, gestures, size) and explain how one sample is turned into a graph.

• Check data quality (missing values, duplicates, outliers, redundant channels) and decide what preprocessing the model needs.

• Check class balance and decide which evaluation metrics are fair.

• Find out whether gestures can be told apart from the features, using statistics, plots, correlation and PCA / t-SNE / UMAP.

• Review GNN-based sEMG papers, compare them in a table, and identify the gap our project can fill.

• Use these findings to define the baseline GNN and the evaluation protocol for the next tasks.

3. Dataset and Graph Representation

3.1 How the data becomes a graph

• Subjects: 40 intact subjects, 3 exercises each (120 files). Exercise 1 has 17 gestures, exercise 2 has 23, exercise 3 has 9: 49 gesture classes in total, plus rest.

• Channels = nodes: 12 surface-EMG electrodes sampled at 2000 Hz. Each electrode is one graph node, so every sample is a graph with 12 nodes. (In the DB2 documentation, 8 electrodes are placed in a ring around the forearm and the other 4 on specific muscle sites; we have not yet used this layout for edges.).

• Windowing: one window = 400 samples = 200 ms. From every gesture repetition (segment) we take one random window, so each subject gives 294 windows and the analysis set has 11,760 windows (40 x 294).

• Node feature = RMS: for each window and channel we compute the root mean square of the signal. RMS is a standard amplitude measure of muscle activation. It turns 200 ms of raw signal

into one number per node, which keeps the graph small and the EDA fast. The cost is that timing and frequency information is discarded; a later model can add richer node features.

3.2 Reason behind using train subjects only in every statistic

Before computing anything, we split the 40 subjects into train / validation / test = 28 / 4 / 8 subjects, with no subject in two groups. Any quantity that is later used to fit a model (scaler, PCA, correlation, graph edges, class weights) is computed on the 28 train subjects only. If the test subjects were included, their information would leak into normalisation and graph construction and the final accuracy would look better than it really is. Only purely descriptive checks (file counts, missing values, duplicates, class counts over all subjects) use all 40 subjects, because nothing is fitted there.

4. Exploratory Data Analysis (EDA)

A. Dataset Summary

Name	NinaPro DB2
Source	ninapro.hevs.ch (Kaggle copy)
Area	Biomedical - sEMG hand gesture recognition
Samples (time steps)	207,713,115 per channel
Windows analysed	11,760 (200 ms each)
Classes	49 gestures (+ rest)
Features	12 channels, RMS per window (numeric, continuous)
Subjects	40
Class balance	1.00x (max/min); rest = 44% of samples
Missing values	0
Duplicate files	0

B. Descriptive statistics (raw samples, train subjects)

Mean and median are about 0 for every channel and the standard deviation is below 0.0005 (it prints as 0.000), because raw EMG is a signed, very low-amplitude signal. The columns that carry information are the range, skewness, kurtosis and outlier share

Statistics of raw samples, TRAIN subjects only (these decide normalisation).

	mean	median	std	min	max	Q1	Q3	skew	kurtosis	outlier_%
ch01	-0.0	0.0	0.0	-0.006	0.006	-0.0	0.0	-0.845	894.481995	18.552
ch02	0.0	0.0	0.0	-0.006	0.006	-0.0	0.0	10.280	5342.629883	16.420
ch03	-0.0	0.0	0.0	-0.002	0.001	-0.0	0.0	-3.104	263.134003	16.700
ch04	-0.0	0.0	0.0	-0.002	0.005	-0.0	0.0	18.552	3374.097900	15.505
ch05	0.0	0.0	0.0	-0.001	0.002	-0.0	0.0	10.617	2127.731934	7.795
ch06	-0.0	0.0	0.0	-0.001	0.001	-0.0	0.0	-0.144	533.565002	8.852
ch07	0.0	0.0	0.0	-0.009	0.006	-0.0	0.0	-2.523	1092.952026	16.829
ch08	0.0	0.0	0.0	-0.005	0.004	-0.0	0.0	-7.336	1133.743042	17.635
ch09	-0.0	-0.0	0.0	-0.003	0.004	-0.0	0.0	5.590	1284.354004	15.629
ch10	-0.0	0.0	0.0	-0.007	0.008	-0.0	0.0	0.616	1543.194946	17.386
ch11	-0.0	-0.0	0.0	-0.002	0.002	-0.0	0.0	3.604	3023.106934	10.210
ch12	-0.0	-0.0	0.0	-0.000	0.001	-0.0	0.0	3.022	134.279007	8.926

Read it: mean ~ 0 (EMG is a signed signal). Kurtosis > 0 = heavy tails (many spikes). Skew ~ 0 = symmetric.

C. Data Quality

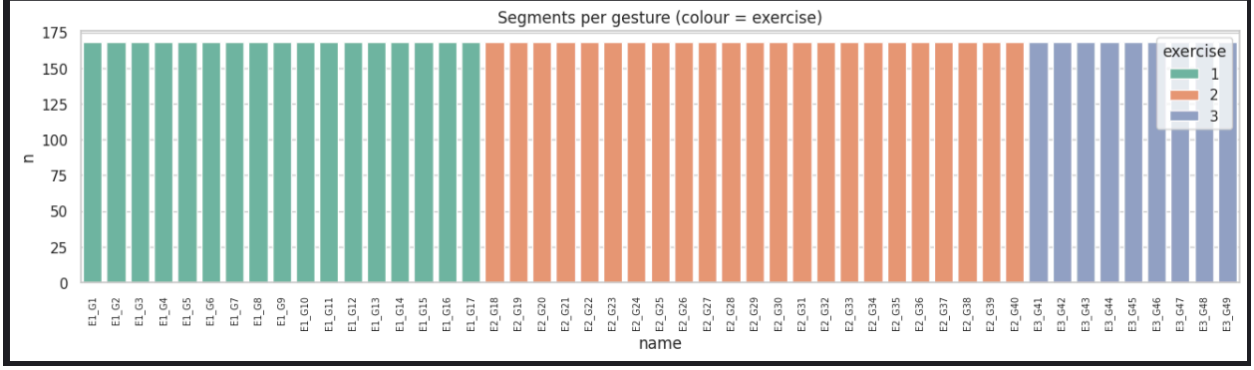
```
Missing values     : 0.0000% [OK]
Duplicate files    : 0 [OK]
Near-constant chan.: none [OK]
Highly correlated  : 0 channel pairs with r > 0.9 [OK]
Outliers (IQR rule): 14.2% of samples
```

	effect on modeling
Missing	Missing values -> need imputation/removal, oth...
Duplicates	Duplicates -> same data in train and test infl...
Near-constant	Near-constant channel -> gives no information,...
High correlation	Highly correlated channels -> redundant inform...
Outliers	Outliers (EMG spikes) -> can dominate scaling;...

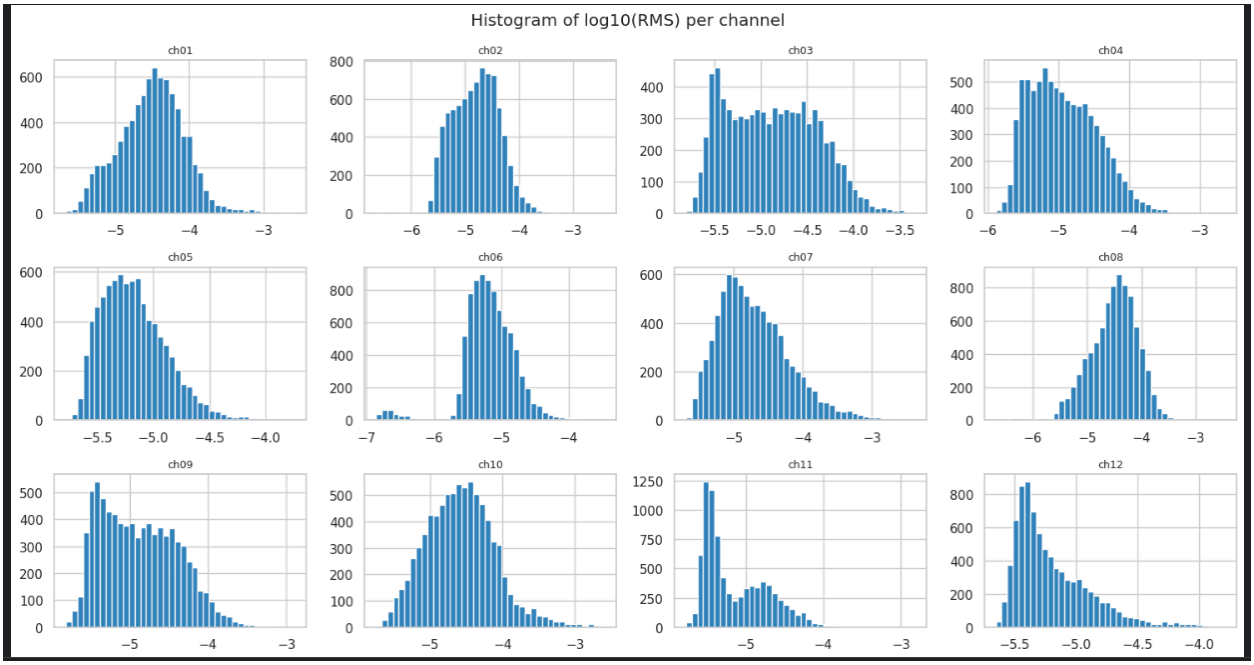
D. Class Balance

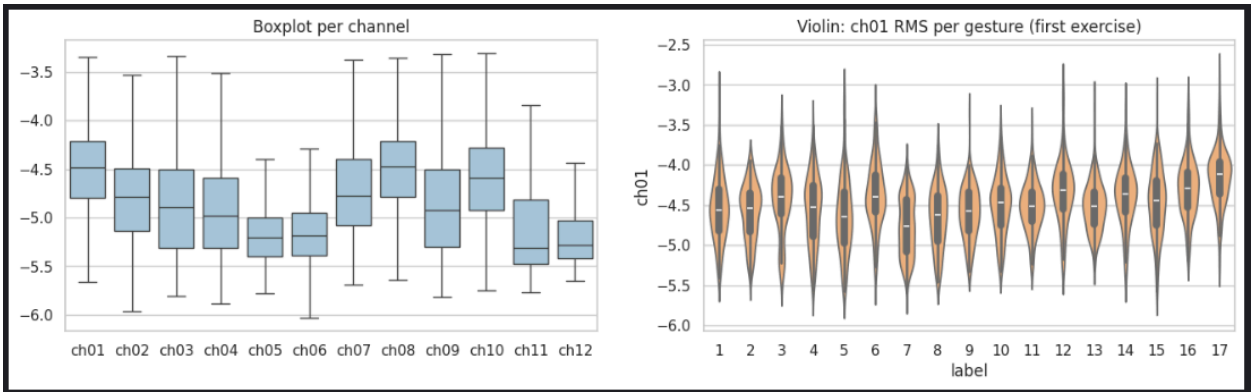
The 49 classes are perfectly balanced: 168 segments per class in the train subjects (28 subjects x 6 repetitions), imbalance 1.00x, and the same 1.00x over all 40 subjects, so the train split is representative. Because of this, plain accuracy is a fair headline metric; we still plan to report macro-F1 and per-class results because 49 classes can hide weak gestures. The rest class is 44% of raw samples, so it is left out of the class counts: if it were included, a model that always predicts rest would score 44% accuracy.

```
TRAIN subjects: imbalance 1.00x | all subjects: 1.00x  -> similar = split is representative
49 gesture classes | segments per class: min 168, max 168 | imbalance 1.00x
Effect on metrics: balanced -> accuracy is a fair metric.
Rest is 44% of raw samples, so it is excluded from the class counts (it would dominate accuracy).
```



E. Feature Distributions

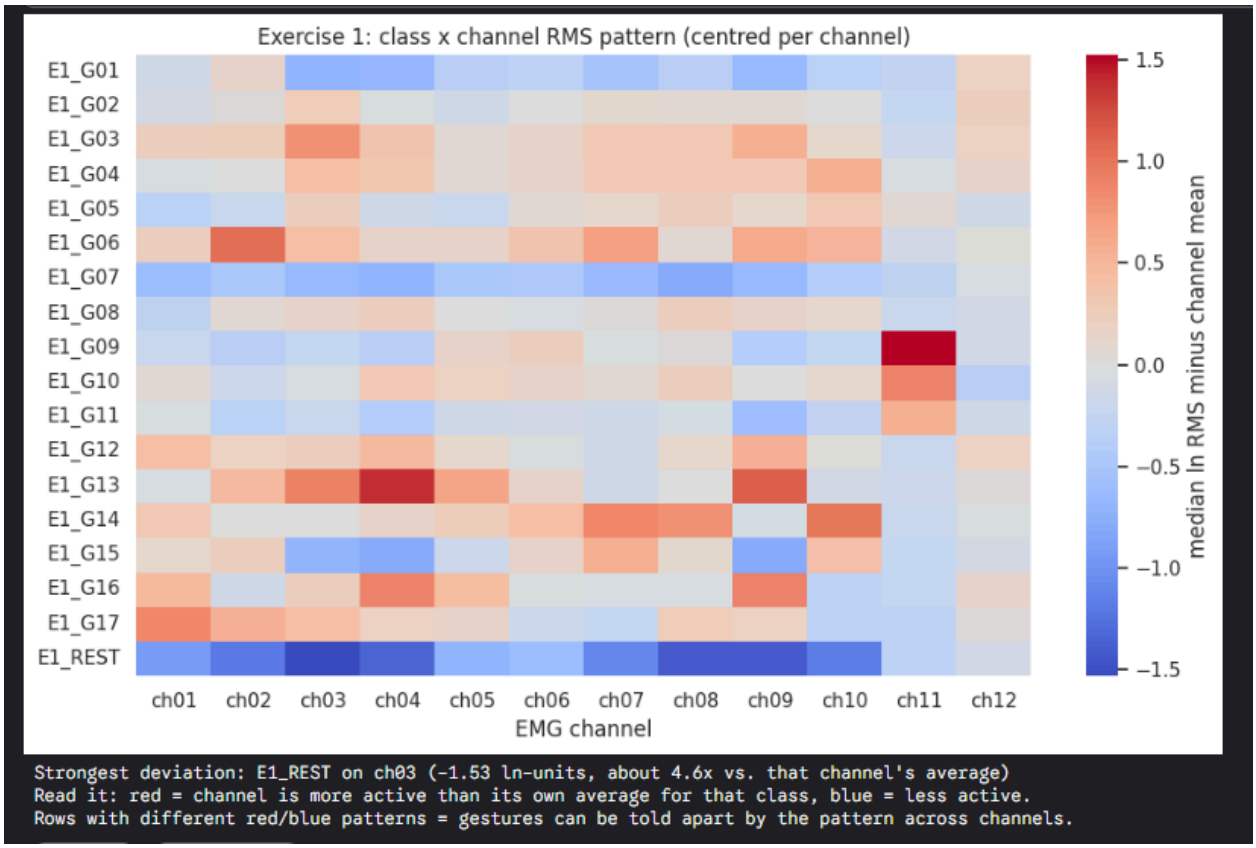




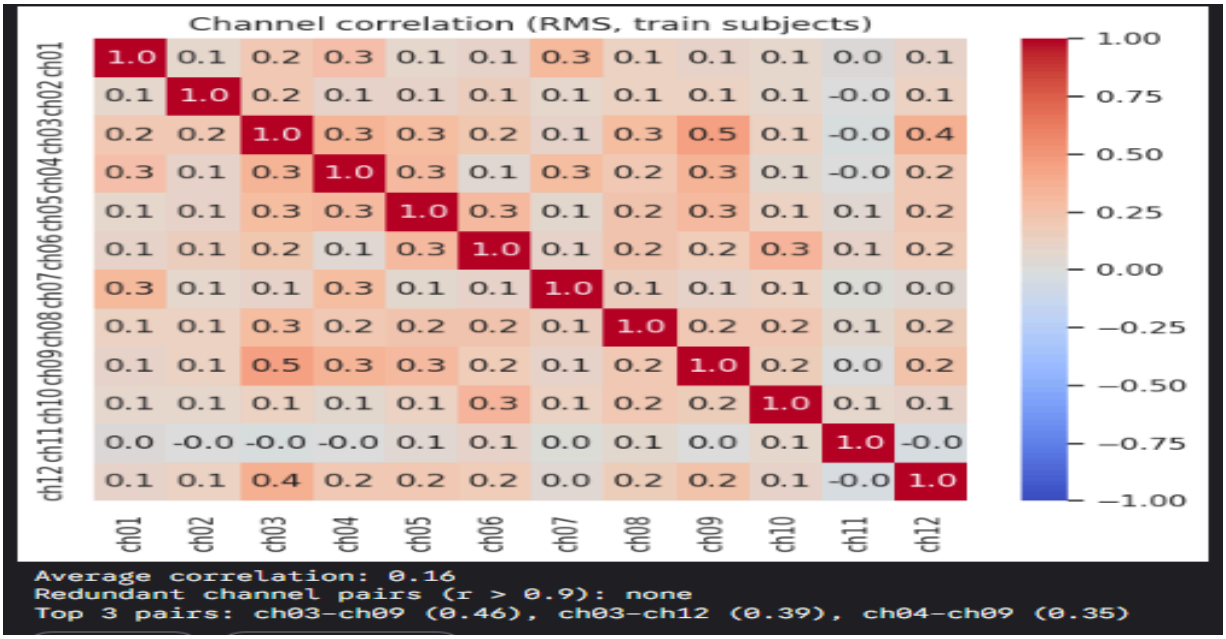

```
Gesture explains 15% of RMS variance on average; best channel ch11: 32%
Read it: if the violins look alike, single-channel RMS cannot separate gestures well; the pattern across channels matters.
```

E2. Class x channel pattern (exercise 1)

To see the cross-channel pattern directly, we took the median ln(RMS) of each class on each channel (exercise 1, train subjects) and subtracted each channel's mean, so red means a channel is more active than its own average for that gesture and blue means less. Rest was added as an extra row.



F. Correlation



G. PCA, t-SNE, UMAP and 3D Visualization

Log-RMS features were standardised with the scaler fitted on train subjects. PCA was also fitted on train subjects; its first three components explain 47%, 12% and 9% of the variance. t-SNE and CSE475 Task 1 | Group 05 | Page 5 UMAP were run on 2,000 train windows of exercise 1 (gestures of different exercises are not directly comparable)



5. Related Work and Gap Analysis

No	Paper	Data and graph	Model and evaluation	Key results	Limitations and gap
1	Vipulanandan et al., Real-Time Gesture Recognition on a GNN on sEMG (arXiv 2026)	Data: Myo band, 8 ch, 8 subjects, 5 gestures. Graph: node = electrode, edge = Pearson correlation per window	Simple 2-layer GNN. Random 20% of overlapping windows as test	98.19% to 99.89%. 48 ms per prediction	only 5 gestures, no cross-subject test; random overlapping split may inflate accuracy. It uses a repetition based split on DB2 and add cross-subject results
2	Hu et al., MFN-SPSCA (IEEE TNSRE 2024)	Data: Myo sEMG and accelerometer; MyoData (17 subj, 8 ch), NinaPro DB5 (10 subj, 16 ch). Graph: node = channel, edge = thresholded Pearson correlation and spatial partitioning	Multi-branch CNN/BiLSTM ,ST-GCN,CWT, cross attention. DB5 reps 1,3,4,6 train, 2,5 test	98.78% MyoData, 91.60% DB5. 12 to 21 ms, about 37 MB	Known users only, no cross-subject; heavy model, needs accelerometer. It used dataset is DB5 not DB2; try a lighter sEMG only GNN with leave-subject out evaluation

3	Xu et al., CNN-MS TGCN (IEEE TNSRE 2024)	Data: own HD sEMG, 11 subj, 17 gestures, 128 ch. Graph: node = electrode, edge = grid distance and learned functional connection	CNN and MSTGCN with spatial temporal attention. User-specific 7:3, user independent 11-fold, transfer learning	96.9% user specific. 68.0% user independent (ResNet50 47.5%, LSTM 57.1%). 92.3% with transfer learning	Offline only, 68% not satisfactory, needs new-user data, no public dataset. It evaluates on public 40 subject DB2, report within subject and cross-subject
4	Lee et al., Spatiotem poral GNNs for Generaliza ble and Interpreta ble EMG Classificat ion (IEEE TNSRE 2025)	Data: BandMyo, NinaPro DB5, UCI, CapgMyo A and B. Graph: node = sensor, edge = physical proximity (KNN 1, KNN-2, fully connected) and temporal links between windows	GCN (also GAT, GraphLSTM, STGCN, GCLSTM). Random 80:20 per subject; intra- and inter-subject	Intra-subject 99.3 to 99.9%. Inter-subject 92.4% DB5, 98.9% UCI, 99.4% BandMyo. 28 sensors give about 97%	Data has been randomly splitted may be optimistic, inter subject protocol unclear, no latency (our note). It do not uses DB2. It uses a clear protocol, compare graph types, analyse channel importance
5	Massa et al., GNNs for HD EMG Movement Intention Recogniti on (IEEE RASSE 2022)	Data: HD-EMG, 20 subj, 65 gestures, 128 ch. Graph: node = channel (65 raw samples), edge = grid distance (128 nodes, 884 edges)	EMG-GNN: 6 SAGEConv + mean pooling + MLP. Each subject separately, 60/20/20 shuffled graphs	Mean error 8.75% (SD 4.92). 45 of 65 gestures under 10% error	The model is weak on gestures with 2+ degrees of freedom; initial study, one dataset, no cross-subject test, no latency. It has a gap on per-class confusion analysis on DB2 and comparison with published baselines
6	Zhou et al., CovGCN (IEEE Sensors Journal 2025)	Data: NinaPro DB2 (40 subj, 12 ch, Ex. B/C/D) and own UOW (8 ch, 3 limb positions). Graph: node = channel, predefined topology refined per gesture by covariance attention	7 graph-conv blocks, 1.07 M parameters. DB2 reps 1,3,4,6 train, 2,5 test, per subject	DB2 Ex. B/C/D: 88.02 / 81.30 / 93.30%. UOW 84.69%. 29.6 ms	Only on specific subject , able-bodied, offline; Transformer slightly higher (88.91% on Ex. B).

6. Research Gap and Planned Baseline

6.1 Overall Summery of Related Works

• Only P6 uses NinaPro DB2, and it evaluates each subject separately. Cross-subject results

on DB2 are missing.
• P1 and P4 use random window splits, which can mix near-identical overlapping windows of the same repetition into train and test (our note). Repetition-based or subject-based splits are more honest.
• The graph is built in very different ways (grid distance, Pearson correlation, covariance, KNN), but no paper compares them under one controlled protocol.
• Latency is reported in only some papers, and per-class or per-channel analysis is rare.

6.2 Our proposed contribution

A reproducible baseline GNN on NinaPro DB2 with repetition-based splits, both within-subject and cross-subject evaluation, and a controlled comparison of graph construction strategies, with per-class analysis of which channels drive each gesture.

6.3 Planned baseline GNN

• Graph per sample: 12 nodes (electrodes), one feature per node (log-RMS, standardised per channel with train statistics), 49-class output.
• Graph variants to compare: (a) fixed anatomical/ring graph, (b) correlation graph from train subjects, (c) learned adjacency. The low channel correlations (mean 0.16) mean a correlation graph needs a deliberate threshold.
• Model: 2 to 3 graph-convolution layers, global mean pooling and a linear classifier; the closest published reference is CovGCN (P6).
• Protocol: subject-wise 28/4/8 split for cross-subject results, and a repetition-based split (as in P6) for within-subject results; report accuracy and macro-F1, and per-class confusion.

6.4 Reason of choosing GNN track

The electrodes form a natural graph; the EDA shows the class information lives in the pattern across channels (Figure 4) rather than in single channels (Figure 3); and the six papers show a clear gap (cross-subject evaluation and graph comparison on DB2).